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Watch Reels & Learn Language

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ΤΜΗΜΑ ΠΛΗΡΟΦΟΡΙΚΗΣ



Πτυχιακή Εργασία

Βλέπε σύντομα βιντεάκια & Μάθε μια Γλώσσα

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Abstract

This thesis presents GloReels, a personalised Mobile-Assisted Language Learning (MALL) application that leverages authentic short-form video content—referred to as Reels—to support second language acquisition. The system addresses three persistent gaps in existing language learning platforms: the absence of authentic, culturally contextualised content; the lack of word-level learner modelling grounded in memory theory; and the prevalence of extrinsic gamification that does not engage active recall.

GloReels is implemented as a three-service microarchitecture: a cross-platform React Native/Expo mobile frontend, a Node.js/Express REST API backend responsible for authentication and vocabulary management, and a Python FastAPI reels microservice that serves linguistically annotated video content. All services share a PostgreSQL 16 relational database whose schema encodes a six-level vocabulary mastery model tracking each learner’s knowledge of individual words alongside a timestamp of last review.

The system draws on three theoretical frameworks. Vygotsky’s Zone of Proximal Development informs the comprehensibility threshold used to filter reels: content is presented only when the learner already knows at least 98% of its vocabulary. The Ebbinghaus forgetting curve provides the basis for a three-stage recommendation pipeline—comprehensibility filtering, spaced-repetition scoring, and exploration blending—designed to surface reels containing vocabulary that is contextually overdue for review. Cognitive Load Theory guides the session design of the two integrated hypercasual games, Wordle and Word of Wonders, whose target words are drawn directly from each learner’s vocabulary model.

The thesis documents the full system architecture, the relational schema, the API design, and the implementation of both game modules. A functional evaluation confirms that the authentication, vocabulary initialisation, reel-serving, and game modules are fully operational. The personalised recommendation pipeline is designed and specified but not yet fully deployed; current reel selection is randomised within the correct learning language.

Limitations, including the absence of a longitudinal user study, are discussed alongside a prioritised roadmap for future work.

Keywords: mobile-assisted language learning, second language acquisition, spaced repetition, authentic materials, personalised recommendation, gamification, React Native

Περίληψη

Εισαγωγή και Κίνητρο

Η εκμάθηση ξένων γλωσσών αποτελεί μια από τις πιο διαδεδομένες ψηφιακές δραστηριότητες στη σύγχρονη εποχή. Η παγκόσμια αγορά εφαρμογών εκμάθησης γλωσσών αξιολογήθηκε στα 5,8 δισεκατομμύρια δολάρια το 2025 και αναμένεται να φτάσει τα 21,1 δισεκατομμύρια δολάρια έως το 2034. Παρά την ταχεία αυτή ανάπτυξη, οι κυρίαρχες πλατφόρμες—όπως το Δουολίνγκο, το Μεμρίσε και το Βυσσυ—παρουσιάζουν κοινές δομικές αδυναμίες: το περιεχόμενό τους είναι αποκομμένο από το αυθεντικό πλαίσιο χρήσης της γλώσσας, η εξατομίκευση περιορίζεται στη ρύθμιση δυσκολίας και η παιχνιδοποίηση δεν ενεργοποιεί την ενεργό ανάκληση λεξιλογίου. Η παρούσα διπλωματική εργασία παρουσιάζει το GloReels, μια εφαρμογή εκμάθησης γλωσσών που αξιοποιεί σύντομα αυθεντικά βίντεο—γνωστά ως Reels—ως βασικό μέσο εκμάθησης, ενισχύοντας την εμπέδωση του λεξιλογίου μέσω εξατομικευμένων υπερτυχαίων παιχνιδιών.

Θεωρητικό Πλαίσιο

Το σύστημα βασίζεται σε τρεις παιδαγωγικές θεωρίες. Η Κοινωνικοπολιτισμική Θεωρία του Vygotsky και η έννοια της Ζώνης Επιχειμένης Ανάπτυξης (ZPD) ορίζουν το κατώφλι κατανόησης: ένα βίντεο επιλέγεται για παρουσίαση μόνο εφόσον ο χρήστης γνωρίζει ήδη τουλάχιστον το 98% του λεξιλογίου του. Η καμπύλη λήθης του Ebbinghaus τεκμηριώνει το μηχανισμό διαστημικής επανάληψης: κάθε λέξη στο μοντέλο λεξιλογίου του χρήστη φέρει επίπεδο κατάκτησης (1–6) και χρονοσήμανση τελευταίας επανάληψης, επιτρέποντας στο σύστημα να προγραμματίζει επανεμφάνσεις πριν επέλθει η λήθη. Η Θεωρία Γνωστικού Φόρτου (CLT) καθοδηγεί τον σχεδιασμό των παιχνιδιών και την εμφάνιση τιτλισμού λέξη-προς-λέξη στα βίντεο, ώστε να μην υπερφορτώνεται το εργαζόμενο μνήμη του χρήστη.

Αρχιτεκτονική Συστήματος

Το GloReels υλοποιείται ως τριμερής μικρο-αρχιτεκτονική υπηρεσιών. Το κινητό frontend αναπτύχθηκε με React Native και Expo SDK 54, επιτρέποντας ανάπτυξη για Android και iOS από έναν κοινό κώδικα σε TypeScript. Το κύριο backend υλοποιήθηκε με Node.js/Express

και διαχειρίζεται την αυθεντικοποίηση (JWT και Google OAuth 2.0), τη διαχείριση προφίλ και τον συγχρονισμό λεξιλογίου. Η υπηρεσία Reels υλοποιήθηκε με Python FastAPI και παρέχει βίντεο εμπλουτισμένα με δομημένα γλωσσικά δεδομένα: διαλόγους, προτάσεις, διακριτικούς κωδικούς ανά λέξη και μεταφράσεις. Τέλος, η βάση δεδομένων PostgreSQL 16 αποθηκεύει το σχεσιακό σχήμα που κωδικοποιεί το μοντέλο λεξιλογίου, τα μεταδεδομένα περιεχομένου και τα σήματα αλληλεπίδρασης χρηστών.

Μοντέλο Λεξιλογίου και Σύστημα Σύγχρονης Ενημέρωσης

Ο πυρήνας του συστήματος είναι ο πίνακας `user_vocabulary`, ο οποίος αντιστοιχίζει κάθε χρήστη σε κάθε λέξη που γνωρίζει, καταγράφοντας επίπεδο κατάρτησης (1–6) και χρόνο τελευταίας επανάληψης. Κατά την εγγραφή, το σύστημα αυτόματα προσθέτει στο μοντέλο του νέου χρήστη όλες τις λέξεις που βρίσκονται κάτω από το δηλωθέν επίπεδο επάρκειάς του, αντιμετωπίζοντας το πρόβλημα της ψυχρής εκκίνησης. Μέσω του μηχανισμού `sync`, ο πελάτης συγκεντρώνει τοπικά τις αλλαγές (εισαγωγές, ενημερώσεις, διαγραφές) και τις αποστέλλει στον διακομιστή σε μια ενιαία αίτηση, επιτρέποντας αδιάλειπτη λειτουργία παιχνιδιών χωρίς σύνδεση στο δίκτυο.

Παιχνιδοποίηση

Το σύστημα ενσωματώνει δύο υπερτυχαία παιχνίδια: Wordle και Word of Wonders. Και τα δύο επιλέγουν τις λέξεις-στόχους απευθείας από το ενεργό μοντέλο λεξιλογίου του χρήστη, δίνοντας προτεραιότητα σε λέξεις με χαμηλό επίπεδο κατάρτησης. Η σωστή ανάκληση αυξάνει το επίπεδο κατάρτησης, ενώ η λανθασμένη το μειώνει, προγραμματίζοντας νωρίτερη επανεμφάνιση. Επιπλέον, ένα σύστημα ενέργειας και νομισμάτων παρέχει βραχυπρόθεσμα κίνητρα για τακτική χρήση.

Αξιολόγηση και Περιορισμοί

Η αξιολόγηση του συστήματος επικεντρώνεται στη λειτουργική πληρότητα και στην ευθυγράμμιση με τους παιδαγωγικούς στόχους. Τα κύρια λειτουργικά στοιχεία—αυθεντικοποίηση, αρχικοποίηση λεξιλογίου, παροχή Reels, παιχνίδια—είναι πλήρως υλοποιημένα. Η εξατομικευμένη μηχανή σύστασης βρίσκεται σε φάση σχεδιασμού, ενώ η τρέχουσα υλοποίηση επιλέγει τυχαία βίντεο εντός της γλώσσας εκμάθησης. Η απουσία εμπειρικής μελέτης χρηστών με μετρήσεις διατήρησης λεξιλογίου αποτελεί τον κύριο περιορισμό της εργασίας.

Μελλοντική Εργασία

Προτεραιότητα αποτελεί η ολοκλήρωση της μηχανής σύστασης (βαθμολόγηση βάσει καμπύλης Ebbinghaus και ανακατεύθυνση εξερεύνησης). Ακολουθούν: αυτοματοποιημένη αναγνώριση ομιλίας για σχολιασμό νέου περιεχομένου, εμπειρική αξιολόγηση μέσω τυχαιοποιημένης ελεγχόμενης μελέτης, επέκταση σε νέες γλώσσες και συνεργατικές λειτουργίες εκμάθησης που

αξιοποιούν την κοινωνική διάσταση της θεωρίας του Vygotsky.

Λέξεις κλειδιά: εκμάθηση γλωσσών με κινητές συσκευές, απόκτηση δεύτερης γλώσσας, διαστημική επανάληψη, αυθεντικό υλικό, εξατομικευμένη σύσταση, παιχνιδιοποίηση, React Native

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I dedicate this work to my family.

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Chapter A

Introduction

A.1 Motivation

The global spread of digital communication has intensified demand for foreign language proficiency, particularly among younger generations who combine professional ambitions with mobile-first lifestyles. Language learning platforms (LLPs) have become the primary vehicle for this demand: the global market reached USD 5.8 billion in 2025 and is projected to grow at a compound annual growth rate (CAGR) of 15.43%, reaching USD 21.1 billion by 2034 [1]. Despite this growth, dominant platforms such as Duolingo, Memrise, and Busuu share a structural limitation: content is largely decontextualised, translation-centred, and disconnected from the authentic, socially situated language that learners encounter in everyday life.

Concurrently, short-form video content—referred to here as Reels—has become the dominant entertainment medium for younger audiences. Platforms such as Instagram, TikTok, and YouTube Shorts command extraordinary daily engagement, yet the language encountered in these videos is consumed passively and leaves no measurable trace in the learner’s long-term memory. The question this thesis addresses is whether these two phenomena—ubiquitous short-form video and mobile language learning—can be productively combined into a system that turns passive viewing into active vocabulary acquisition.

This thesis proposes **GloReels**, a cross-platform mobile application that addresses the identified gaps by embedding vocabulary acquisition within short-form authentic video content and reinforcing it through personalised hypercasual games. The system tracks

each learner’s vocabulary state at the word level, uses that state to select contextually appropriate reels, and adapts game difficulty to the same vocabulary model.

A.2 Background

A.2.1 Short-Form Video

Short-form video content has come to dominate online attention. Generation Z—the primary demographic for language learning applications—allocates a disproportionate share of screen time to Reels and similar formats [11]. Research indicates that more than half of this cohort watches long-form video content on social media platforms, suggesting a sustained appetite for video-based information delivery [7]. However, high engagement does not translate to retention: without active processing, vocabulary encountered in video dissolves rapidly following the forgetting curve described by Hermann Ebbinghaus in the 19th century. Unless a word is deliberately reviewed at increasing intervals, memory of it decays exponentially.

A.2.2 Hypercasual Games

Hypercasual games—defined by their minimal ramp-up time, single-mechanic loops, and session lengths suited to short commutes—represent a similarly underutilised learning medium. Their accessibility and habitual engagement patterns make them natural candidates for spaced-repetition delivery. GloReels integrates two such games, Wordle and Word of Wonders, both of which select their target words directly from the learner’s active vocabulary model, creating a closed loop between video exposure and game-based reinforcement.

A.3 Problem Statement and Research Questions

The central research problem is the absence of a language learning system that simultaneously leverages (i) the authentic contextual richness of user-generated short-form video, (ii) word-level learner modelling grounded in established memory theory, and (iii) gamified reinforcement adapted to the individual learner’s vocabulary state.

This thesis addresses the following research questions:

- RQ1.** How can a mobile application integrate authentic short-form video content with a word-level vocabulary tracking model to support second language acquisition?
- RQ2.** How should a content recommendation engine be designed to select reels that are both comprehensible and lexically challenging for a given learner, based on the Ebbinghaus forgetting curve and CEFR proficiency levels?
- RQ3.** To what extent can hypercasual game mechanics, when parameterised by the learner’s vocabulary state, serve as effective spaced-repetition exercises within a language learning application?

A.4 Significance

The significance of this work is threefold. First, it addresses a demonstrated gap in current LLPs by providing authentic audiovisual content supported by vocabulary scaffolding, rather than isolated translation drills. Second, it operationalises established learning theories—Vygotsky’s Zone of Proximal Development, Ebbinghaus spaced repetition, and Cognitive Load Theory—within a concrete, deployable mobile architecture. Third, the system is designed for mobile-first use: it supports offline gameplay, cross-platform deployment via React Native, and a gamification layer (energy, coins, experience) that maintains engagement over time.

A.5 Thesis Structure

The remainder of this thesis is organised as follows. Chapter B reviews mobile-assisted language learning platforms, identifies their limitations, and establishes the pedagogical framework. Chapter C presents the system architecture, technology stack, data model, and recommendation engine design. Chapter D documents the implementation of each component with reference to the source code and database schema. Chapter E evaluates the system against its design objectives and discusses performance. Chapter F states the contributions, acknowledges limitations, and proposes future work.

Chapter B

Literature Review

B.1 Mobile-Assisted Language Learning

Digital language learning platforms (LLPs) have become dominant in foreign language education, with over 70 applications serving millions of users through Mobile-Assisted Language Learning (MALL) [6]. The global market size for these platforms was valued at USD 5.8 billion in 2025, projected to grow to USD 21.1 billion by 2034 at a CAGR of 15.43% [1]. This rapid expansion has been further accelerated by Large Language Models (LLMs): Duolingo—which commands over 50% of global downloads [12]—required 12 years to create its first 100 language courses, yet added 148 new courses in early 2025 alone, citing AI-driven content generation as the primary enabler [3].

B.2 Limitations of Current Platforms

Karasimos [6] conducted a systematic cross-platform review of five representative LLPs—Duolingo, Memrise, Busuu, Rosetta Stone, and LingQ—against a set of pedagogical and usability criteria. While all five platforms demonstrate strengths in structured vocabulary delivery and habit formation, the review reveals consistent gaps that motivate the present work.¹

¹Throughout this chapter, “current language learning platforms” refers to the five applications examined by Karasimos [6], unless otherwise specified.

B.2.1 Uniform Content Regardless of Learner Demographics

All five platforms present uniform content regardless of user demographics (children, Generation Z, adults, elderly), personal interests, or sociocultural context. Duolingo employs machine learning algorithms to adjust content difficulty, but personalisation is limited to the difficulty dimension alone [2]; topic, domain, and cultural framing remain fixed.

B.2.2 Insufficient Multimedia Integration

Duolingo’s primary instructional mode is text and audio built around translation exercises. This approach does not exploit visual memory—a well-documented channel for vocabulary retention—nor does it expose learners to the prosodic and paralinguistic cues present in natural speech.

B.2.3 Lack of Authentic, Real-World Content

Most platforms—Duolingo, Busuu, Memrise, and Rosetta Stone—lack authentic, real-world materials in their instructional design. LingQ partially integrates such content; however, as noted by Karasimos [6], it is not recommended as a starting point for beginners. Gilmore [5] demonstrates that authentic materials are crucial for developing communicative competence and for sustaining learner motivation. At the same time, Liu and Dong [8] caution that authentic language materials can create comprehension difficulties for learners who lack the cultural background knowledge assumed by native-speaker communication. In GloReels, this tension is addressed by pairing authentic video content with cultural annotations and adaptive vocabulary scaffolding, ensuring that exposure to real-world language does not exceed the learner’s Zone of Proximal Development (ZPD).

B.2.4 Shallow Gamification Mechanics

The gamification elements present in most LLPs—streaks, point tallies, and leaderboards—do not constitute genuine game design. They are motivational overlays applied to repetitive translation drills. This distinction matters because research in game-based learning distinguishes between extrinsic reward structures, which are fragile and dependency-forming, and intrinsic challenge loops, which sustain engagement through meaningful problem-

solving [9].

B.2.5 Lack of Support for Advanced Proficiency Development

With the exception of LingQ, no platform in Karasimos’s review provides adequate support for learners at B2 level and above. Advanced learners require exposure to authentic idiomatic usage, domain-specific vocabulary, and complex syntactic structures—precisely the content available in user-generated short-form video but absent from structured drills.

B.2.6 Absence of Sociocultural Context

Language is inseparable from culture. The five reviewed platforms treat vocabulary and grammar as context-free systems, ignoring the sociocultural theory articulated by Vygotsky, which holds that language development is fundamentally mediated by social interaction and cultural context.

B.3 Pedagogical Foundations

B.3.1 Vygotsky’s Sociocultural Theory

Lev Vygotsky’s Sociocultural Theory [14] holds that cognitive development arises through social interaction mediated by cultural tools, of which language is the primary instrument. Central to the theory is the concept of the **Zone of Proximal Development (ZPD)**: the distance between what a learner can accomplish independently and what they can achieve under guidance or in collaboration. Effective instruction operates within the ZPD—content that is too easy provides no developmental stimulus, while content that is too difficult cannot be processed without first establishing prerequisite knowledge.

In GloReels, the ZPD is operationalised through the comprehensibility threshold of the recommendation engine. A reel is considered suitable for a given learner when that learner already knows at least 98% of its vocabulary—a threshold informed by Nation’s [10] research on the vocabulary coverage required for unassisted reading comprehension. The remaining 2% constitutes the “i+1” challenge (one step beyond the learner’s current knowledge), which falls squarely within the ZPD.

B.3.2 Spaced Repetition and Memory Retention

Hermann Ebbinghaus’s empirical research demonstrated that memory retention decays exponentially over time in the absence of review [4]. The resulting **forgetting curve** underpins all modern spaced-repetition systems (SRS): by scheduling review of a word at increasing intervals timed to precede the moment of forgetting, the system converts short-term exposure into durable long-term memory.

GloReels implements a six-level mastery model in the `user_vocabulary` table. Each word entry records a `mastery_level` (1–6) and a `last_review` timestamp. The recommendation engine uses these values to prioritise reels containing words whose review is overdue, ensuring that vocabulary resurfaces at intervals calibrated to each learner’s individual retention rate.

B.3.3 Cognitive Load Theory

Cognitive Load Theory (CLT), introduced by Sweller [13], distinguishes between *intrinsic load* (the inherent complexity of the material), *extraneous load* (demands imposed by poor instructional design), and *germane load* (cognitive effort invested in schema formation). Effective instruction minimises extraneous load while preserving germane load.

In GloReels, CLT informs two design decisions. First, each reel is accompanied by timestamped sentence-level captions and per-token translations, allowing learners to direct their attention to the target vocabulary without being overwhelmed by unknown words. Second, game sessions are capped and paced through an energy system, preventing fatigue-induced cognitive overload.

B.4 Short-Form Media in Second Language Acquisition

Authentic audiovisual materials have long been recognised as valuable in Second Language Acquisition (SLA) research. Short-form video, in particular, provides a dense multimodal signal: spoken language, prosodic variation, facial expression, gesture, and cultural context are all co-present, reinforcing one another in ways that text-based instruction cannot replicate.

The challenge is retention. Without active processing, vocabulary encountered during

video viewing follows the Ebbinghaus forgetting curve as surely as any other exposure. GloReels addresses this by treating each reel not as a standalone learning event but as a contextualised encounter with target vocabulary in the learner’s personal vocabulary model. Words seen in a reel are subsequently reinforced in game sessions, and the recommendation engine surfaces those words again in future reels, creating a multi-context learning loop that supports durable acquisition.

B.5 Gamification in Education

Genuine gamification—as opposed to the point-and-streak overlays critiqued in Section 2.2.4—involves embedding core game mechanics into the learning activity itself, such that the challenge of the game is inseparable from the learning objective [9]. Hypercasual games are well suited to this purpose: their short session length, immediate feedback loops, and escalating difficulty can be parametrically linked to a learner’s vocabulary model.

GloReels integrates two hypercasual game formats: a Wordle variant, in which the target word is drawn from the learner’s vocabulary model, and Word of Wonders, a grid-based word-construction game whose level vocabulary is similarly personalised. In both cases, success requires active recall rather than passive recognition, engaging a fundamentally different and more durable memory process than translation drills.

Chapter C

System Architecture and Methodology

C.1 System Overview

GloReels is designed as a three-service architecture in which a mobile frontend, a primary REST API backend, and a dedicated reels microservice each handle a well-defined subset of the system’s responsibilities. Figure ?? illustrates the high-level interaction between these components. All services share a single PostgreSQL database instance and are co-ordinated in development via Docker Compose.

- **Frontend:** A cross-platform mobile application built with React Native (Expo SDK 54), targeting Android and iOS from a single TypeScript codebase.
- **Backend API:** A Node.js/Express service (ESM modules, port 3500) responsible for user authentication, vocabulary management, progress synchronisation, and the dictionary.
- **Reels Service:** A Python FastAPI microservice responsible for serving reels enriched with structured dialogue data—sentences, timestamped tokens, and translations.
- **Database:** PostgreSQL 16, whose schema encodes the vocabulary model, user progress, content metadata, and engagement signals.

C.2 Technology Stack

C.2.1 Frontend — React Native / Expo

React Native was selected to enable cross-platform deployment from a single codebase without sacrificing the native rendering performance required for smooth video playback. Expo SDK 54 provides managed native modules for video (`expo-video`), secure credential storage (`expo-secure-store`), and build tooling (`eas-cli`). The application uses Expo Router (file-system-based navigation) and React Native Reanimated for gesture-driven animations in the game interfaces.

C.2.2 Backend — Node.js / Express

The backend is implemented as an ES-module Node.js application using Express 4. This choice reflects the project’s existing expertise in JavaScript and the maturity of the Express ecosystem for REST API development. Authentication is handled with JSON Web Tokens (JWT): a short-lived access token (15 minutes) and a long-lived refresh token (30 days) follow the standard rotation pattern. Google OAuth 2.0 is supported alongside email/password authentication via the `google-auth-library`. Input validation uses Joi schemas for every endpoint, and the API is documented with Swagger (OpenAPI 3.0) accessible at `/swagger`.

C.2.3 Reels Service — Python / FastAPI

The reels microservice is implemented in Python using FastAPI and SQLAlchemy (async). The separation from the main backend is motivated by the richer data-processing requirements of the reels pipeline: joining reels with their associated dialogues, sentences, per-token word data, and sentence translations in a single query is better expressed in SQLAlchemy’s ORM than in raw SQL via the Node.js `pg` client.

C.2.4 Database — PostgreSQL 16

PostgreSQL was chosen for its support of complex relational queries, array operators, and procedural functions—all of which are used in the vocabulary bulk-initialisation and the planned recommendation engine. The full schema is maintained in `database/thesis_db.sql`.

C.3 Data Modelling

The database schema models the domain along three axes: linguistic content, user state, and engagement signals.

Linguistic content is organised hierarchically. A **word** entry stores the written form, part of speech, CEFR level (N, A1, A2, B1, B2, C1, C2), optional article, audio URL, and image URL. Words are grouped into **sentences** via the **sentence_tokens** join table, which also records the token’s position and part of speech in context. Sentences are grouped into **dialogues** via **dialogue_sentences**, which adds per-sentence timestamps (**start_time_ms**, **end_time_ms**) to enable synchronised subtitle rendering. Each **reel** references one dialogue, tying video content to its full linguistic annotation.

User state is captured in two tables. **user_languages** records each language pair a user is studying, their CEFR proficiency level, and accumulated experience points. **user_vocabulary** is the core of the learner model: each row binds a user to a word and records a **mastery_level** (1–6) and a **last_review** timestamp. The mastery scale encodes qualitative stages from initial encounter (1) through active recall (6), and it is updated both by game outcomes and by reel interaction events.

Engagement signals are stored in **reel_interactions**, which records views, likes, saves, comments, shares, and reports per (user, reel) pair. These signals are reserved for the personalised recommendation pipeline described in Section C.4.

C.4 Recommendation Engine Design

The recommendation engine is the theoretical core of the system. Its goal is to select, from the pool of available reels, those that are simultaneously *comprehensible* (the learner knows enough vocabulary to follow the content) and *pedagogically productive* (the reel contains words that are due for review or represent the next developmental step). The design follows a three-stage pipeline.

C.4.1 Stage 1: Comprehensibility Filtering

For each candidate reel, the engine computes the proportion of the reel’s unique word tokens that are present in the learner’s vocabulary model at mastery level ≥ 1 . A reel

passes the filter only if this proportion meets or exceeds 98%, corresponding to Nation’s [10] threshold for unassisted reading comprehension. This criterion implements the ZPD operationally: the learner encounters real language while remaining within a comprehensible range.

C.4.2 Stage 2: Spaced-Repetition Scoring

Each reel is scored according to the review urgency of the vocabulary it contains. For each word w in the reel with mastery level m and days since last review d , the system computes an urgency score based on the Ebbinghaus forgetting function. The reel’s aggregate score is the mean urgency across its target vocabulary. This stage ensures that reels containing words the learner is about to forget are ranked above reels containing words that were recently reviewed.

C.4.3 Stage 3: Re-Ranking and Exploration

The top-ranked candidates from Stage 2 are blended with a random sample of unseen reels to prevent the recommendation feed from collapsing into a narrow subset of the content library. This exploration component is essential for exposing learners to new vocabulary and cultural contexts, broadening their ZPD over time.

Implementation status: The current release implements Stage 1 as a CEFR-level pre-filter applied at registration (words below the user’s stated proficiency are bulk-added to the vocabulary model at mastery level 3). Stages 2 and 3 are designed and specified but not yet fully implemented in the reels service; the service currently returns randomised reels filtered by learning language. Full implementation of the scoring pipeline is identified as future work in Chapter F.

C.5 Gamification Design

The gamification layer consists of two components: a progression system and two vocabulary-driven games.

The **progression system** awards experience points (XP) for completed game rounds and reel interactions, advancing the user’s CEFR proficiency level in `user_languages.experience`.

Alongside XP, an energy mechanism limits the number of game rounds per session, preventing fatigue-induced cognitive overload consistent with Cognitive Load Theory. Coins provide a secondary reward currency redeemable for in-app benefits.

The two games—**Wordle** and **Word of Wonders**—share the same word selection principle: each game round draws its target vocabulary from the learner’s active vocabulary model, prioritising words that are due for spaced-repetition review. Correct recall triggers a mastery-level increment in `user_vocabulary`; incorrect attempts either maintain or decrement the mastery level, scheduling the word for earlier re-exposure. This closes the loop between reel-based exposure and active game-based recall.

Chapter D

Implementation Details

D.1 Database Schema

The database schema (`database/thesis.db.sql`) defines fourteen tables across three domains. The core vocabulary tables are reproduced schematically in Table D.1.

Table	Key columns
words	id, written_form, part_of_speech, level (CEFR), language_id, audio_url, ima
user_vocabulary	user_id, word_id, user_languages_id, mastery_level (1–6), last_review
user_languages	user_id, native_language_id, learning_language_id, proficiency_level, exper
reels	id, url, thumbnail_url, dialogue_id, language_id, duration
sentence_tokens	sentence_id, word_id, position, part_of_speech

Table D.1: *Core tables of the GloReels PostgreSQL schema.*

The `user_vocabulary.mastery_level` column is constrained to the range 1–6 via a `CHECK` constraint. A database trigger function (`update_last_login_on_refresh_token`) automatically records the `last_login` timestamp whenever the refresh token is updated, removing the need to handle this in application code.

D.2 Reels Service

D.2.1 Service Architecture

The reels microservice (`reels-service/`) is a FastAPI application backed by an async SQLAlchemy session. Its single primary endpoint, `GET /api/v1/reels`, accepts three parameters: `native_language_code`, `learning_language_code`, and `limit` (1–50, default 10). The endpoint validates that the two language codes are distinct, resolves them to database language IDs, and delegates to `ReelService.get_random_reels()` (`app/services/reel_servi`

D.2.2 Dialogue Enrichment

Each reel response embeds the full dialogue associated with the reel’s video content. The `build_dialogue_response()` method in `reel_service.py` assembles this structure via a single SQLAlchemy query that eager-loads `DialogueSentence` $\rightarrow$ `Sentence` $\rightarrow$ `SentenceToken` $\rightarrow$ `Word` in one round trip, ordered by sentence position. For each sentence, a separate query resolves the native-language translation via `sentence_translations`. The resulting response carries per-token word data (written form, CEFR level, part of speech, audio URL, image URL) alongside sentence-level timestamps, enabling the front-end to render synchronised word-level subtitles.

D.2.3 Personalisation Implementation Status

The `GET /api/v1/reels` endpoint includes an optional Bearer token header. When a valid JWT is present, the router is designed to invoke a personalised recommendation method; in the current release this path falls through to the same random-selection logic as unauthenticated requests (see `reels.py` lines 87–98). Full personalisation via the scoring pipeline described in Chapter C is planned for a future release.

D.3 Backend API

D.3.1 Authentication Module

The backend (`backend/`) implements a dual-token authentication scheme. On successful login or registration, the server issues a short-lived JWT access token (default: 15 minutes, configurable via `ACCESS_TOKEN_EXPIRES_IN`) and a long-lived refresh token (default: 30 days). The refresh token is persisted in the `users.refresh_token` column and returned to the client as an HTTP-only cookie. The `GET /api/v1/refresh` endpoint validates the cookie, verifies the stored token, and issues a new access token without requiring the user to re-authenticate.

Google OAuth 2.0 is supported via the `google-auth-library` package. The `POST /api/v1/auth/google` endpoint verifies the Google ID token using `OAuth2Client.verifyIdToken()`, extracts the user profile, and either retrieves an existing account or creates a new one, then issues the same token pair.

D.3.2 Registration and Vocabulary Initialisation

The registration endpoint (`controllers/registerController.js`) performs the following operations atomically:

1. Validates the request body against a Joi schema (`validation/RegisterSchema.js`).
2. Checks for duplicate email addresses.
3. Hashes the password using `bcrypt`.
4. Creates the user record with up to three username variants derived from the email prefix.
5. Creates the `user_languages` record for the selected language pair.
6. Bulk-inserts all `words` whose CEFR level is strictly below the user's stated proficiency into `user_vocabulary` at mastery level 3 ("understood"), using a single parameterised `INSERT ... SELECT` query against the `words` table.
7. Merges any additional vocabulary changes submitted during onboarding.

Step 6 is implemented in `models/userVocabularyModel.js` as the `addByProficiencyLevel()` method. The CEFR levels are ordered as `['N', 'A1', 'A2', 'B1', 'B2', 'C1', 'C2']`; the method slices all levels below the user's stated level and passes them as a PostgreSQL array parameter to the `ANY($5)` clause, avoiding a variable-length `IN` clause.

D.3.3 Vocabulary Synchronisation

The `PATCH /api/v1/user/sync` endpoint (`controllers/syncController.js`) provides an offline-first synchronisation mechanism. The client accumulates vocabulary changes locally and pushes them as a single batch: inserts (new word encounters), updates (mastery level changes following game outcomes), and deletes (words removed from the model). The server processes deletes first, then updates, then inserts, ensuring that concurrent local changes do not create phantom records.

D.4 Game Modules

D.4.1 Word of Wonders

Word of Wonders is a grid-based word-construction game in which the player must identify hidden words from a set of provided letters. Each game level is associated with a set of target words. In *GloReels*, these target words are drawn from the player's active vocabulary model, prioritising words at lower mastery levels. The grid is generated by computing all valid arrangements of the target words' letters into a crossword-style grid.

D.4.2 Wordle

The Wordle implementation presents the player with a fixed number of attempts to guess a hidden word of known length. Each guess is evaluated for exact matches (correct letter, correct position) and partial matches (correct letter, wrong position), with colour feedback consistent with the original game's conventions. The target word is selected from the player's vocabulary model in the same manner as Word of Wonders.

Both games post vocabulary state updates to the sync endpoint upon session completion, propagating mastery changes to the server and closing the reinforcement loop.

D.5 Complexity Analysis

Vocabulary initialisation (`addByProficiencyLevel`): The dominant operation is a single `INSERT ... SELECT` query whose cost is $O(|W_L|)$, where $|W_L|$ is the number of words in the learning language at levels below the user’s proficiency. For a language with a vocabulary of n words, this is bounded by $O(n)$.

Reel retrieval (`get_random_reels`): The query performs a filtered table scan on `reels` ($O(|R_L|)$, where $|R_L|$ is the number of reels in the learning language), followed by a `func.random()` sort ($O(|R_L| \log |R_L|)$) and a `LIMIT`. For each returned reel, the dialogue enrichment requires $O(|D|)$ where $|D|$ is the number of tokens in the dialogue. In practice, $|R_L|$ is small relative to full-text indices, so the dominant cost is the join graph traversal.

Vocabulary synchronisation: The sync operation processes inserts, updates, and deletes in serial loops within a single request. For a batch of k changes, the cost is $O(k)$ database round trips in the worst case. For large offline sessions this could become significant; batching all updates into a single parameterised statement is identified as a performance improvement for future work.

Chapter E

Evaluation and Results

E.1 Evaluation Approach

A full empirical user study—with longitudinal vocabulary retention measurements, control groups, and statistical analysis—is beyond the scope of this undergraduate thesis. The evaluation presented in this chapter is therefore structured at two levels: (i) a functional analysis of the implemented system components against the design objectives stated in Chapter C, and (ii) a heuristic assessment of the system’s alignment with the theoretical framework established in Chapter B. Where quantitative data could be collected from the running system, it is reported; where it could not, the design rationale is evaluated against established criteria.

E.2 Functional Evaluation

Table E.1 maps each design objective from Chapter C to its implementation status and the evidence for that status.

E.3 Performance Analysis

The following measurements were taken against the local Docker Compose stack on a standard development machine.

Design objective	Status	Evidence
User registration with proficiency-based vocabulary initialisation	Implemented	<code>registerController.js</code> : bulk INSERT via <code>addByProficiencyLevel()</code>
JWT authentication with refresh token rotation	Implemented	<code>routes/auth.js</code> , <code>routes/refresh.js</code>
Google OAuth 2.0 login	Implemented	<code>controllers/googleAuthController.js</code>
Reel serving with dialogue, sentence tokens, and translations	Implemented	<code>reels-service/app/services/reel.service.py</code>
Offline-capable vocabulary synchronisation	Implemented	<code>controllers/syncController.js</code>
Wordle game with vocabulary-driven word selection	Implemented	<code>frontend/components/games/Wordle/</code>
Word of Wonders game with vocabulary-driven level generation	Implemented	<code>frontend/components/games/WordOfWonders/</code>
Energy and coins gamification system	Implemented	<code>users.energy</code> , <code>users.coins</code> columns; PATCH <code>/api/v1/users/me/energy</code>
Personalised recommendation (Ebbinghaus scoring)	Partial	CEFR-level pre-filter implemented; spaced-repetition scoring not yet deployed

Table E.1: *Functional evaluation of GloReels against design objectives.*

Reel retrieval latency: The `GET /api/v1/reels` endpoint with `limit=10` returned responses in the range of 80–150 ms on a populated database containing 50 reels and their associated dialogue data. The dominant cost was the dialogue enrichment join (Section 4.2.2); caching the dialogue structure in a materialised view or an in-memory cache would reduce this significantly at scale.

Registration with vocabulary initialisation: For a language containing approximately 500 words at CEFR levels N through A2, the bulk INSERT in `addByProficiencyLevel()`

completed in under 20 ms. This confirms the $O(n)$ complexity estimate from Section 4.5.

Vocabulary synchronisation: A batch of 30 vocabulary updates processed by `syncController.js` completed in under 50 ms. As noted in Section 4.5, the serial per-word UPDATE loop is the scalability bottleneck for large offline sessions.

E.4 Pedagogical Alignment

Zone of Proximal Development. The comprehensibility threshold implemented via CEFR-level filtering at registration ensures that the vocabulary model contains only words the learner has declared they know or have been exposed to. This operationalises the lower bound of the ZPD. The upper bound—the “i+1” challenge—is present structurally in every reel that contains target vocabulary at one CEFR level above the learner’s current proficiency, though the automated selection of such reels awaits the full recommendation pipeline.

Ebbinghaus spaced repetition. The `mastery_level` and `last_review` fields in `user_vocabulary` provide the data substrate for a spaced-repetition scheduler. Game sessions currently increment or decrement mastery level based on correct or incorrect recall, and the sync mechanism ensures these changes persist. The scheduling function that would use these values to prioritise review-due words in reel selection is the principal missing component.

Authentic content. The reel-serving architecture delivers genuine video content annotated with timestamped, token-level linguistic data. The design supports cultural annotations and adaptive scaffolding at the data-model level (dialogue sentences can carry free-text explanatory notes), though this UI affordance is not yet exposed in the current frontend.

Gamification quality. Both games draw their target vocabulary from the learner’s personal vocabulary model rather than a fixed word list. This satisfies the criterion for genuine gamification (Section 2.5) in that the challenge of the game is inseparable from the learner’s individual vocabulary state.

E.5 Limitations

The primary limitation of this evaluation is the absence of a longitudinal user study. The claims in Section 5.4 are design-level arguments grounded in the theoretical framework; they do not constitute empirical evidence that GloReels improves vocabulary retention relative to a control condition. A randomised controlled study with pre- and post-tests measuring vocabulary gain, retention at one week and one month, and learner engagement would be required to substantiate these claims.

A secondary limitation is that the personalised recommendation pipeline is not fully implemented. The current system serves random reels within the correct language, which satisfies the authenticity criterion but not the spaced-repetition criterion. The evaluation therefore cannot assess the system’s behaviour as a recommendation engine.

Chapter F

Conclusions and Future Work

F.1 Contributions

This thesis has presented GloReels, a personalised mobile language learning application that combines authentic short-form video content with a word-level vocabulary tracking model and gamified reinforcement. The following contributions were made:

1. **A three-service system architecture** comprising a React Native/Expo mobile frontend, a Node.js/Express REST API backend, and a Python FastAPI reels microservice, all sharing a PostgreSQL 16 database. The architecture is containerised with Docker Compose and production-ready in its authentication and data management components.
2. **A relational vocabulary model** that tracks each learner’s mastery of individual words across a six-level scale, recording mastery level and last-review timestamp per (user, word, language pair) tuple. This model provides the data substrate for spaced-repetition scheduling and CEFR-based comprehensibility filtering.
3. **A proficiency-aware registration pipeline** that bulk-initialises a new user’s vocabulary model by pre-populating all words below their stated CEFR level, eliminating the cold-start problem that affects systems relying on incremental word encounters.
4. **An offline-first vocabulary synchronisation mechanism** (POST /api/v1/user/sync)

that batches client-side mastery updates and propagates them to the server in a single round trip, enabling continued game-based reinforcement without network connectivity.

5. **Two vocabulary-driven hypercasual games** (Wordle and Word of Wonders) whose target words are drawn from the learner’s active vocabulary model, creating a closed loop between reel-based contextual exposure and active game-based recall.
6. **A design specification for a three-stage recommendation engine** (comprehensibility filtering, Ebbinghaus scoring, exploration blending), grounded in Vygotsky’s ZPD and the forgetting curve, which serves as the roadmap for future development.

F.2 Limitations

The work presented in this thesis has several limitations that must be acknowledged honestly.

Incomplete recommendation pipeline. The personalized recommendation engine described in Chapter C is designed and specified but not fully implemented. The current release serves randomly selected reels within the correct learning language. The system therefore demonstrates the architecture for personalised delivery but does not yet realise its learning-theoretic benefits.

No empirical user study. The pedagogical effectiveness of the system has not been evaluated in a controlled experiment. The evaluation in Chapter E is design-level and heuristic. Whether GloReels produces measurable vocabulary gains relative to a control condition is an open empirical question.

Limited content library. The system requires a curated library of reels with complete linguistic annotation (tokenisation, CEFR-level word tagging, timestamped sentence alignment, and translation). Building and maintaining such a library at scale requires either significant manual curation effort or an automated annotation pipeline that is outside the scope of this work.

Single-platform gamification. The current energy and coins system incentivises engagement at a surface level. Whether these mechanics sustain long-term vocabulary prac-

tice—rather than producing the “drilling masquerading as gamification” pattern identified in Chapter B—has not been tested.

F.3 Future Work

The most immediate priority for future work is completing the recommendation pipeline. Stage 2 (Ebbinghaus-based review urgency scoring) requires implementing the scheduling function that computes review priority from `mastery_level` and `last_review`, and integrating it into `reel_service.py` as the currently commented-out `get_personalized_reels()` method. Stage 3 (exploration blending) is architecturally straightforward once Stage 2 is in place.

Beyond the recommendation engine, the following directions are identified as high-value:

- **Automated content annotation pipeline:** A pipeline that ingests a raw video URL, transcribes speech using a large language model, aligns the transcript to video timestamps, tokenises and POS-tags the output, maps tokens to the word database, and produces a fully annotated reel record—reducing the per-reel curation cost from hours to minutes.
- **Empirical evaluation:** A randomised controlled study measuring vocabulary gain, retention at one week and one month, and learner engagement metrics, comparing GloReels against a no-game control condition and against Duolingo.
- **Multi-language expansion:** The schema and service layer are language-agnostic (language is a foreign key throughout); adding a new language requires only content ingestion, not code changes.
- **Collaborative learning features:** Shared vocabulary challenges, leaderboards scoped to friend groups, and comment-driven vocabulary discussion on individual reels, supporting the social dimension of Vygotsky’s theory.
- **Vocabulary synchronisation performance:** Replacing the serial per-word UPDATE loop in `syncController.js` with a bulk `INSERT ... ON CONFLICT DO UPDATE` statement to support large offline sessions without latency degradation.

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Abbreviations

AI — Artificial Intelligence

API — Application Programming Interface

CAGR — Compound Annual Growth Rate

CEFR — Common European Framework of Reference for Languages

CLT — Cognitive Load Theory

CORS — Cross-Origin Resource Sharing

ESM — ECMAScript Modules

HTTP — Hypertext Transfer Protocol

JWT — JSON Web Token

LLM — Large Language Model

LLP — Language Learning Platform

MALL — Mobile-Assisted Language Learning

OAuth — Open Authorisation

ORM — Object-Relational Mapping

PLLA — Personalised Language Learning Application

POS — Part of Speech

REST — Representational State Transfer

SLA — Second Language Acquisition

SRS — Spaced-Repetition System

UI — User Interface

URL — Uniform Resource Locator

UX — User Experience

ZPD — Zone of Proximal Development

Glossary

Authentic materials — Language learning resources drawn from real-world communicative situations, produced by native speakers for native speakers rather than created artificially for instructional purposes.

CEFR level — A proficiency descriptor from the Common European Framework of Reference for Languages, ranging from A1 (beginner) through C2 (mastery). Used in GloReels to tag words and to initialise the learner’s vocabulary model.

Cognitive Load Theory — An instructional design theory that distinguishes between intrinsic load (complexity of the material), extraneous load (demands from poor design), and germane load (effort directed at schema formation). Effective instruction minimises extraneous load.

Dialogue — In the GloReels schema, a structured sequence of sentences associated with a reel, carrying per-sentence video timestamps that enable synchronised subtitle rendering.

Ebbinghaus forgetting curve — An empirical model of memory decay showing that retention drops exponentially over time without review. The basis for spaced-repetition scheduling systems.

Energy system — A gamification mechanism in GloReels that limits the number of game rounds per session, preventing cognitive overload and encouraging regular short sessions rather than infrequent long ones.

Hypercasual game — A game genre characterised by a single core mechanic, minimal learning curve, and session lengths of one to five minutes, suited to mobile use during short breaks.

Mastery level — A value from 1 to 6 stored in `user_vocabulary` that represents a learner’s degree of familiarity with an individual word, from initial exposure (1) to durable

active recall (6).

Microservice — An architectural pattern in which a system is decomposed into small, independently deployable services, each responsible for a specific domain. GloReels uses this pattern to separate the reels pipeline from the main backend API.

Reel — A short-form video clip, typically under 90 seconds, drawn from a social media platform or user-generated content. In GloReels, each reel is annotated with a full linguistic dialogue.

Scaffolding — Temporary instructional support provided to a learner to enable them to accomplish a task they could not yet perform independently, progressively removed as competence develops.

Spaced repetition — A learning technique in which review of material is scheduled at increasing time intervals, calibrated to the moment immediately before forgetting occurs, to consolidate long-term retention.

Token — In computational linguistics, a single word or punctuation unit identified within a sentence. In the GloReels schema, `sentence_tokens` links each sentence to its constituent words with position and part-of-speech information.

Zone of Proximal Development (ZPD) — Vygotsky’s concept of the gap between what a learner can do independently and what they can achieve with guidance. Effective instruction targets this zone. In GloReels, the 98% vocabulary comprehensibility threshold is the operational proxy for the ZPD lower bound.